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First-Order Predicate Logic

Chapter Learning Outcomes

CLO

3

- Understand the meaning of a predicate.
- Learning syntax and semantics of predicate logic.
- Translate an English sentence to FOPL and vice-versa.
- Know and understand "for all" and "there exists" quantifiers.
- Write the negation of statements with quantifiers "for all" and "there exists".
- Learn about nested quantifiers.

Predicate Logic

Are the following sentences, propositions?

1. X + Y = 10.
2. X is the instructor of our AI course.

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Predicate Logic

A **Predicate** is an assertion that contains one or more variables.

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Predicate Logic

In Propositional Logic

Ali is the father of Maha.

P: Ali is the father of Maha

In Predicate Logic

Ali is the father of Maha.

father_of(Ali, Maha)

Predicate Logic

Also called “First-Order Logic”

Why we need Predicate Logic

- We need to be able to refer to **objects**.
- We need to refer to **relations between objects**.
- If we can refer to objects, we also want to be able to **capture the meaning of every and some of**.

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Predicate Logic : Syntax

The syntax of predicate logic consists of:

- Truth Symbols
- Constants
- Variables x, y, …
- Functions f(), g(), …..
- Predicates p(), q(), …
- Logical connectives $\wedge$, $\vee$, $\neg$, $\rightarrow$, $\leftrightarrow$
- Quantifiers $\forall$, $\exists$

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Predicate Logic : Syntax

1. **Truth symbols** true and false (these are reserved symbols)

2. **Constant symbols** are symbol expressions having the first uppercase (Stand for objects). E.g., Today, Laith

3. **Variable symbols** are symbol expressions beginning with an lowercase character. E.g., x, y, z

4. **Function symbols** are symbol expressions having the first character lowercase (Stand for functions).

Arity : number of elements in the domain

E.g., mother-of (Bill); maximum-of (7,8)

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Symbols and Terms

- A function expression consists of a function constant of arity n, followed by n terms, t_1 ,t_2 ,…, t_n, enclosed in parentheses and separated by commas.
  - ✓ mother-of(mother-of(Joe))
  - ✓ maximum(maximum(7, 18), add_one(18))

- A term is either a constant, variable, or function expression.
  - ✓ color_of(house_of(neighbor(Joe)))
  - ✓ house_of(x)

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Predicates vs. Functions

Predicate

Predicate is a relation for which is more than one answer and then choosing the correct one.

- teacher(Hasan, Ahmad).
- less_than(4, 6).

Function

Function is a relation for which is one answer.

- sqrt(9).
- distance(Amman, Irbid).

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Translating English Sentences to Predicate Logic

Translating sentences without variables or quantifiers is straightforward. Just remember that proper names translate into individual constants and verbs, common nouns, and adjectives into predicates.

Ali is a student.

student(Ali)

Translating English Sentences to Predicate Logic

EXAMPLE

Ali is the father of Maha and Salma.

In Propositional Logic

In Predicate Logic

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Quantifiers

Universal Quantifier: ∀x

Asserts that a sentence is true for all values of variable x.

∀x: loves(x, AI)

Existential Quantifier: ∃x

Asserts that a sentence is true for at least one value of a variable x .

∃x: loves(x, AI)

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Quantifiers

TABLE 1 Quantifiers.

Statement	*When True?*	*When False?*
$\forall x P(x)$	$P(x)$ is true for every x.	There is an x for which $P(x)$ is false.
$\exists x P(x)$	There is an x for which $P(x)$ is true.	$P(x)$ is false for every x.

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How to Write Logical Connectives Using Keyboard

For All

@ (at symbol shift with 2)

How to Write Logical Connectives Using Keyboard

There Exists

 # (hash symbol shift with 3)

Translating English Sentences to FOPL

EXAMPLE

Some students are sick today.

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Translating English Sentences to FOPL

EXAMPLE

All students are sick today.

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Summary

	Symbol	Main Connective
Universal Quantification	$\forall$	$\rightarrow$
Existential Quantifier	$\exists$	$\wedge$

Universal Quantifier: $\forall x$

Asserts that a sentence is true for all values of variable x.

$\forall x$: loves(x, Fish)

Existential Quantifier: $\exists x$

Asserts that a sentence is true for at least one value of a variable x .

$\exists x$: loves(x, Fish)

Translating English Sentences to FOPL

EXAMPLE

All basketball players are tall.

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Translating English Sentences to FOPL

EXAMPLE

Some books are interesting

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Translating English Sentences to FOPL

EXAMPLE

Nobody likes taxes

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Translating English Sentences to FOPL

EXAMPLE

Someone is sleeping

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Translating English Sentences to FOPL

EXAMPLE

No one is sleeping

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Translating English Sentences to FOPL

EXAMPLE

Everyone is sleeping

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Translating English Sentences to FOPL

EXAMPLE

Not Everyone is sleeping

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General Identities

$$\forall x\ \neg p(x) \equiv \neg \exists x\ p(x)$$

$$\neg \forall x\ p(x) \equiv \exists x\ \neg p(x)$$

$$\forall x\ p(x) \equiv \neg \exists x\ \neg p(x)$$

$$\exists x\ p(x) \equiv \neg \forall x\ \neg p(x)$$

Nested Quantifiers

Nested quantifiers are quantifiers that occur within the scope of other quantifiers.

EXAMPLE

Everybody loves somebody

Nested Quantifiers

EXAMPLE

Everybody loves everybody

$$\forall x\ \forall y : person(x) \wedge person(y) \rightarrow loves(x, y)$$

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Nested Quantifiers

EXAMPLE

Everybody loves himself or herself.

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Nested Quantifiers

EXAMPLE

There is a student who has passed all AI exams.

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Negating Quantified Expressions

$$\forall x : p(x) \rightarrow q(x)$$

$$\Big\downarrow \text{negation}$$

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Negating Quantified Expressions

$$\exists x : p(x) \wedge q(x)$$

$$\Big\downarrow \text{negation}$$

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Negating Quantified Expressions

EXAMPLE

Translate the following sentence to FOPL, write down the negated expression and then translate the negated expression in English.

No cats like Pepsi

Negating Quantified Expressions

EXAMPLE

Translate the following sentence to FOPL, write down the negated expression and then translate the negated expression in English.

Some happy people are healthy and wealthy

Nesting Quantifiers

- **The order of quantifiers of the same type doesn't matter.**
  - $\forall x\ \forall y$: **loves(x, y)** : says everyone love everyone.
  - $\forall y\ \forall x$: **loves(x, y)** : says everyone is loved by everyone.
- **The order of mixed quantifiers does matter:**
  - $\exists x\ \forall y$: **loves(x, y)** : Says there is a person who loves everyone in the world.
  - $\forall y\ \exists x$: **loves(x, y)** : Says everyone in the world is loved by at least one person.

Nesting Quantifiers

EXAMPLE

- $\forall x \exists y$: mother(x, y)

Everyone has a mother

(**correct**)

- $\exists y \forall x$: mother(x, y)

There is a person who is the mother of everyone

(**wrong**)

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Translating English Sentences to FOPL

EXAMPLE

All employees with salary is greater than 1000JD, pay taxes.

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Translating English Sentences to FOPL

EXAMPLE

All of Bill’s kids are also Hillary's kids

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Translating English Sentences to FOPL

EXAMPLE

There exist a kid whose father is Bill and whose mother is Hillary

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Translating English Sentences to FOPL

EXAMPLE

All books written by Dickens are interesting

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Translating English Sentences to FOPL

EXAMPLE

No test is difficult

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Translating English Sentences to FOPL

EXAMPLE

There are movies that are boring

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Translating English Sentences to FOPL

EXAMPLE

All S are P

Some S are P

Some S are not P

No S are P

Translating English Sentences to FOPL

EXAMPLE

Anybody that has a friend is not lonely

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Translating English Sentences to FOPL

EXAMPLE

If a person is female and is a parent, then this person is someone's mother.

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Translating English Sentences to FOPL

EXAMPLE

Everything attracts everything.

Everything is attracted by everything

Something attracts something

Translating English Sentences to FOPL

EXAMPLE

Something is attracted by something

Nothing attracts anything

Nothing is attracted by anything

Translating English Sentences to FOPL

EXAMPLE

Everything attracts something

Something is attracted by everything

Everything is attracted by something

Translating English Sentences to FOPL

EXAMPLE

Something attracts everything

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Check - Chapter Learning Outcomes

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End of Chapter

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